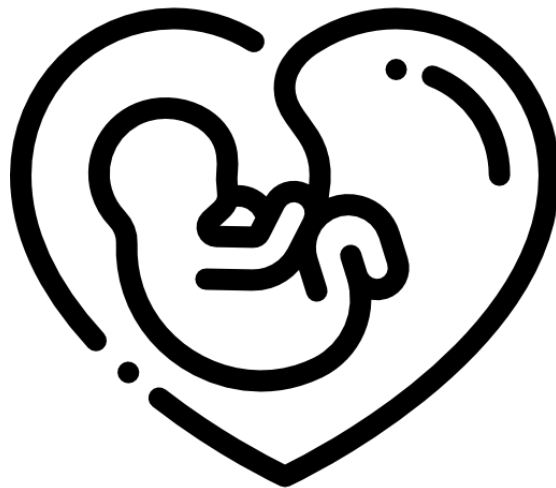




**Mount Sinai
Hospital**

Sinai Health System
Joseph & Wolf Lebovic
Health Complex
Leadership Sinai Centre for Diabetes

Gestational Diabetes Mellitus (GDM) and Pregnancy



Being told you have GDM can be a shock and upsetting.

**This booklet will provide you with some information about
how you can have a healthy baby!**

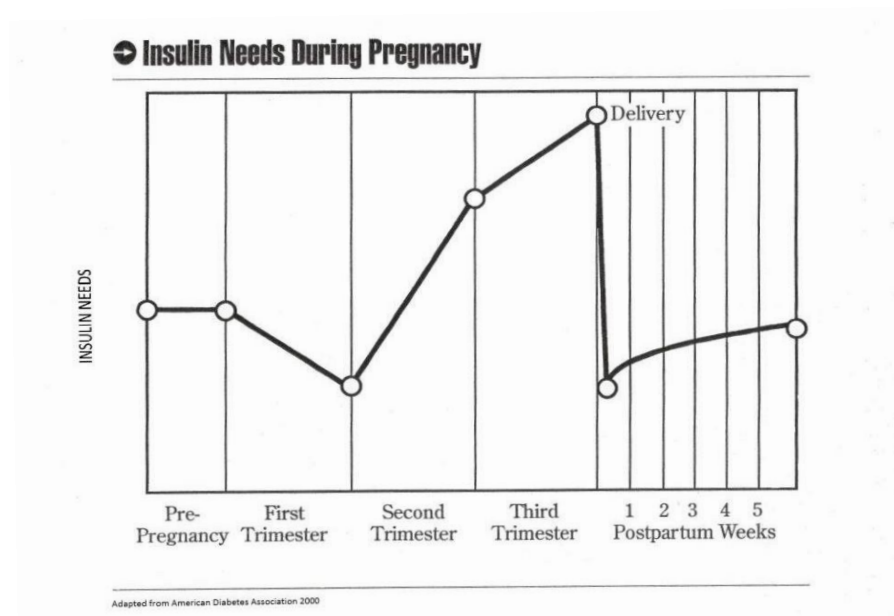
What is Gestational Diabetes?

Gestational Diabetes Mellitus (GDM) means your blood sugars are higher than normal in pregnancy.

Why does this happen?

- Your body needs food for energy. When you eat, your body changes some of the food (carbohydrates) into sugar (glucose). In order for the body to use the sugar for energy it needs insulin. Insulin is a hormone made by the pancreas which helps to move the sugar from your blood into the body cells to use as energy.
- During the second half of pregnancy (24-28 weeks), the placenta (tissue that feeds your baby) makes hormones that are necessary for the growth of your baby.
- These hormones interfere with how the insulin in your body works and this causes blood sugars to increase. The pancreas will make more insulin to bring down the blood sugar.
- GDM develops when the pancreas cannot make enough insulin to keep up with the pregnancy hormones.

The picture below shows you how your insulin needs change during pregnancy:



At about 6 weeks – 3 months after delivery your insulin needs will be almost back to your normal levels. It is very important to do the 2 hour glucose tolerance test between 6 weeks to 3 months after delivery to make sure that your insulin needs have gone back to normal and you do not have diabetes.

Risk Factors and Effects of GDM

How would you know if you are more likely to develop GDM?

GDM does not have anything to do with what you did or did not do or eat; **you did not cause gestational diabetes**. The rates of GDM are increasing worldwide – up to 20% of pregnant people develop gestational diabetes, depending on their risk factors.

The following factors increase your risk of getting GDM:

Being:

- 35 years of age or older
- From a high-risk group (African, Arab, Asian, Hispanic, Indigenous, or South Asian)

Having:

- Obesity (a body mass index greater than or equal to 30 kg/m²)
- Prediabetes
- Gestational diabetes in a previous pregnancy
- Given birth to a baby that weighed more than 4 kg
- A parent or sibling with type 2 diabetes
- Polycystic ovarian syndrome (PCOS) or acanthosis nigricans (darkened patches of skin)

Using:

- Corticosteroid Medication (e.g. prednisone, celestone)

Note: Some people may need to receive Celestone injections during their pregnancy. Please let your endocrinologist know if you need Celestone as it does affect your blood glucose levels.

How was it decided that you have GDM?

- Gestational diabetes is diagnosed by doing either a glucose challenge test (GCT), which is not fasting, and/or a 75g oral glucose tolerance test (OGTT), for which you have to fast overnight.
- If one of the values is not normal then you will be diagnosed with gestational diabetes.
- It means that after drinking the sweet drink, your pancreas was not able to make enough insulin to bring your blood sugar back to normal values within the time it normally takes.

How does GDM affect the baby?

Your baby will not have diabetes because you have gestational diabetes.

During pregnancy your baby gets their nourishment from you. If your blood sugar levels are high then your baby will get extra sugar from you.

- The baby will make extra insulin to lower its blood sugar; this gives the baby extra energy/calories, which allows the baby to grow larger.
- A large baby may mean a difficult delivery for both you and your baby.

After delivery the baby's blood sugar level can drop too low (this is called hypoglycemia, when a baby's sugar <2.8 mmol/L) for a short time because the baby's pancreas is slow at turning off the extra insulin it was used to making.

- The nurse will test the baby's blood sugar using a heel prick. If the blood sugar is low, you will be asked to breastfeed or your baby will be given formula. Your baby's blood sugar will go back to normal.
- Having a large baby may cause a difficult delivery, shoulder dystocia (an infant's shoulder becoming lodged in the birther's pelvis).
- Your baby may have jaundice.
- Research has shown that babies born to people who had gestational diabetes are at greater risk of becoming obese as young adults. As they grow older they are at greater risk of developing impaired glucose tolerance (IGT) or prediabetes.

How does GDM affect you?

- You may have increased risk of a caesarean section (C-Section)
- You may have high blood pressure or preeclampsia
- There is an increased risk of having gestational diabetes with next pregnancy
- There is an increased risk of developing type 2 diabetes in future
- There is an increased risk of developing cardiovascular disease in the future

What can I do to manage Gestational Diabetes?

1. Healthy eating

Use the GDM Meal Planner

Below are some general tips about healthy meals. **Please follow dietitian's recommendations.**

- Meal planning is the first step because it provides necessary nutrition and can control both weight gain and blood sugar levels for you and your baby
- You need to have carbohydrates at every meal and snack.
 - Do not cut out carbohydrates from your meal plan to control your blood sugar. Your body will start to break down fat for energy instead of the food you eat, and will begin to produce ketones which can be harmful to your baby.
- Keep notes of the foods you eat (including portion size i.e. 125ml or ½ cup). This will show you how the different types and amounts of food affect your blood sugar. It will also help the dietitian so they can work with you to make any needed changes.

2. Activity

If you have any obstetrical or medical conditions please do not exercise until you have discussed this with your doctor.

- Any activity that gets you moving will help you better manage your blood sugar.
- Moderate activity can lower your blood sugar (Moderate activity = you are slightly breathless and have a slight increase in your heart rate e.g. walking). A great way to increase your activity is to start walking for 15-20 minutes or swimming. Then gradually increase up to at least 150 minutes weekly of moderate activity.
- Try not to nap after your meal(s) as you will likely have higher blood sugars.
- Regular activities will also keep you fit and prepare you for the birth of your baby.

3. Checking your blood sugars

- **Always wash your hands before testing.**
- If you get a result that does not make sense, wash your hands and repeat your blood sugar test. However, generally try to only perform one test per time slot.
- Checking your blood sugars helps you and your health care team to see if the changes to food and activity are working, and to make adjustments as needed.

NOTE:

The blood sugar targets in pregnancy are much lower than targets for type 1 or type 2 diabetes.

4. Take insulin, if required

When to Check Your Blood Sugar

1. When you **wake up in the morning (before breakfast)**
2. Check **1 hour after the start/first bite of your meal** (breakfast/lunch/dinner)
 - E.g. if you start eating lunch at 12:00pm then you should check at 1:00pm for your 1 hour after lunch test (eat in ~20-30 mins).

NOTE: You do have the option to occasionally check 2 hour after the **start/first bite of food**. Please ensure you indicate that sugars were tested at the 2 hour mark.

Blood sugar targets during pregnancy	
Before breakfast 1 st thing in the morning <i>(At least 8hrs or more without food)</i>	Less than or equal to 5.2 mmol/L
1 hour after each meal Breakfast, lunch, supper	Less than or equal to 7.7 mmol/L
2 hours after each meal Breakfast, lunch, supper	Less than or equal to 6.6 mmol/L

Remember: You should avoid eating and drinking in the 1 hour before checking your blood sugar levels. You can drink water at any time as it will not affect your blood sugar.

What do you need?

1. Finger poker and lancets (dispose of lancets in sharps container only)
2. Test strips
3. Blood sugar meter
4. Logbook or blood sugar diary

Test Strips without Insurance

If you do not have private insurance coverage, ask your Endocrinologist for a form from Diabetes Canada which will cover about 75% of the cost of the test strips. This is called the Monitoring for Health Program.

- Your pharmacy may want you to pay the full cost 100% and then you have to apply to the Diabetes Canada for 75% of the cost of the strips. Remember to save all of your receipts.

Note:

If at any time your blood sugar numbers are higher than the targets 3-4 times in one week at the same time, please contact your doctor or diabetes educator.

e.g. 3-4 out of 7 fasting (before breakfast) blood sugars are high:

DAY	S	M	T	W	T	F	S
Before Breakfast	In target	High	High	High	In Target	High	In Target

When To Use Insulin in Pregnancy

Many people with gestational diabetes will not be able to keep their blood sugars in target range with the GDM meal plan and activity. Insulin will help bring the blood sugars into target range until you deliver your baby.

Remember:

Insulin is safe in pregnancy, as it *does not* cross the placenta to the baby.

Your endocrinologist may recommend that you need to take insulin to manage your blood sugars. To start using insulin:

1. your endocrinologist will give you a prescription
2. you will attend the insulin start class to learn about insulin and how to inject properly

Insulin Start Classes

In Person Class**Wednesdays, 2:00 pm**

Small group appointment at the LSCD clinic:
60 Murray Street, 5th Floor

- All patients and staff wearing masks
- Wear comfortable two piece clothing for the injection demonstration

Virtual Appointment**Thursdays, 2:00 pm**

Small group appointment via Zoom

- Keep an eye out for appointment information via email that day
- Insulin prescription sent to your pharmacy for pick up before class
- Consent form required before booking
- Wear comfortable two piece clothing for the injection demonstration

What Happens After the Baby is Born?

Will the gestational diabetes go away? Yes

- ✓ Once the baby is born and the placenta is delivered, GDM should go away. The day after delivery the nurse will draw a blood sample before and 2 hours after breakfast to make sure that your blood sugar has returned to normal.
- ✓ The next important test is the 2 hour glucose tolerance blood test around 3 months after delivery that will make absolutely sure that you do not have diabetes.

Remember:

- ✓ People with gestational diabetes are at increased risk of developing diabetes, pre-diabetes conditions, and cholesterol problems later in life.
- ✓ Remind your family doctor that you had GDM so that you can have your blood sugar and your cholesterol levels checked as required.

How can I decrease my risk of developing type 2 diabetes?

- Target a **healthy weight** to reduce your risk of gestational diabetes in the next pregnancy and developing type 2 diabetes
- Eat **balanced meals** with lots of vegetables.
- **Exercise regularly** such as walking for ~30 minutes 5 days per week (at least 150 min per week) at a moderate pace has shown to lower the chance of developing type 2 diabetes.
- If you choose to breastfeed it is all the nutrition baby will need in the first 6 months of life.
 - **Breastfeed** immediately after birth and for a minimum of 4 months in order to prevent hypoglycaemia (low blood sugar) in your newborn, and lower the risk for obesity in childhood, and diabetes for both you and your child.
 - You can still raise a healthy and happy baby with formula.

What if I want to become pregnant again?

Before: Be screened for type 2 diabetes after your pregnancy (6-12 weeks) AND before planning another pregnancy

When pregnant: All pregnant people with previous GDM should be screened for gestational diabetes between 24 to 28 weeks or earlier if at risk for type 2 diabetes.

Remember: We know that if you get pregnant and your blood sugar levels are very high, it can be very harmful during the early development of your baby.

- Start following the meal plan that you are using now as soon as you decide **or** find out you are pregnant. See your doctor who will order either the non-fasting glucose test (GCT) or the fasting glucose test OGTT.

Further GDM Information

Please be careful when obtaining information from the internet, not all the sites available will provide you with accurate, factual information. The list below contains credible sites that can provide you with more information on gestational diabetes and pregnancy.

Diabetes Canada www.diabetes.ca

American Diabetes Association <https://diabetes.org>

Association of Diabetes Care & Education Specialists <https://www.adces.org>

Mother to Baby <https://mothertobaby.org>

The Organization of Teratology Information Specialists (OTIS) 1-866-626-6847

References:

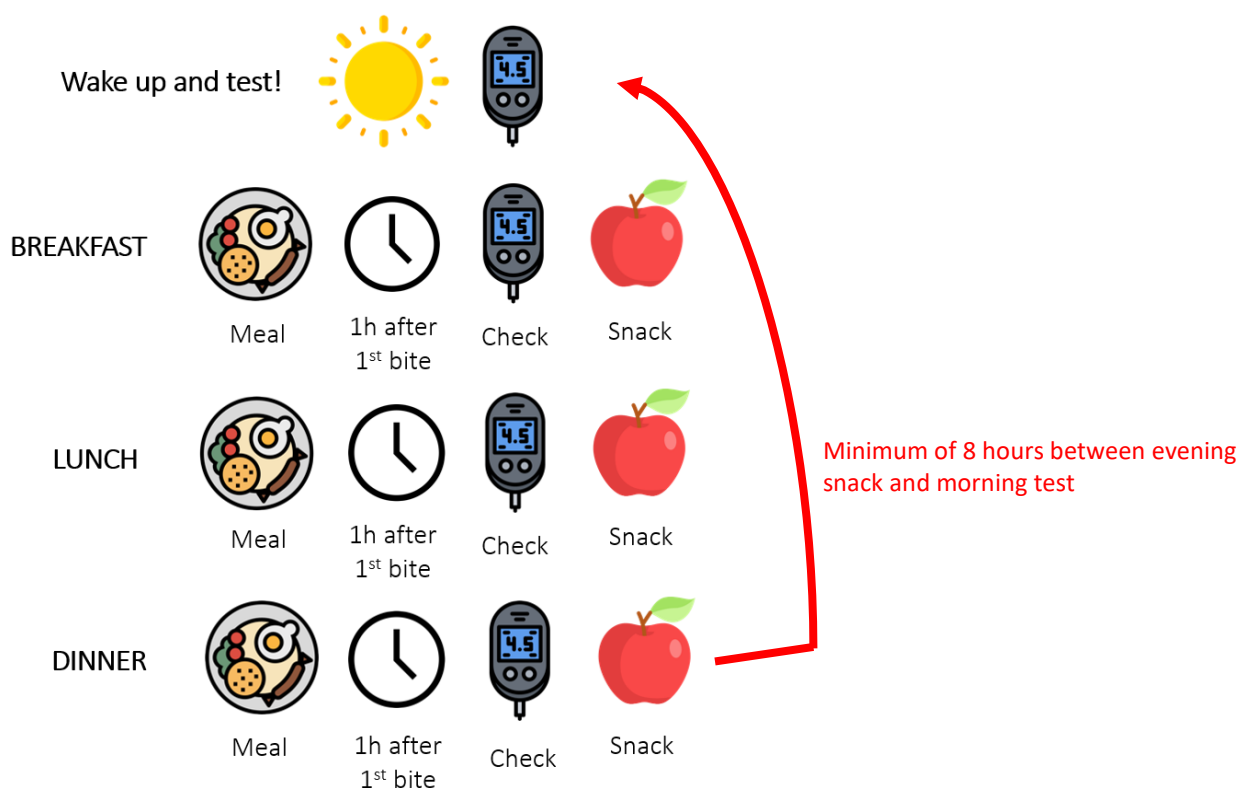
Feig DS, Berger H, Donovan L. et al. *Diabetes Canada 2018 Clinical Practice Guidelines for the Prevention and Management of Diabetes in Canada: Diabetes and Pregnancy*. Can J Diabetes 2018; 42 (Suppl 1):S255-S282.

Summary of Key Points

1. Be active every day (play often – increase activity e.g. 20-30 min walks)
2. Eat 3 meals and 2-3 snacks and follow the GDM meal plan
3. Avoid having carbohydrate foods/drinks between your meal and your 1 hour blood sugar test
4. Start with checking blood sugars 4 times a day every day for now:
 - a. Before breakfast, AND 1 hour after breakfast, lunch and supper/dinner
 - b. Always wash your hands before checkingFurther instruction will be provided on the follow up with your diabetes doctor next week.
5. Always bring your meter and log book with you to all your appointments
6. **Always call your doctor/diabetes educator if 3-4 blood sugar results are higher than normal at the same time (e.g. 3-4 out of 7 readings after dinner are high).**

Typical GDM Day

Icons from Freepik, Nikita Golubev, photo3idea_studio, Those Icons



Program Steps

1. **Attend initial visit with endocrinologist** – in-person (Wednesdays)
2. **Attend group Introduction to Gestational Diabetes Class** – virtual on zoom (Thursday mornings)
3. **Submit blood glucose records** – prior to dietitian phone call
4. **Have individual follow-up with dietitian** – by phone (Mondays and some Tuesdays)
5. **Attend follow-up with endocrinologist** – in-person or by phone (Wednesdays)
As needed:
6. **Attend Insulin Start Class** – in-person (Wednesday afternoons) or virtual (Thursday afternoons)
7. **Submit blood glucose records** – prior to dietitian or nurse phone call
8. **Have individual follow-up with dietitian or nurse** – by phone (as scheduled)
9. **Attend follow-up with endocrinologist** – in-person or by phone (Wednesdays)

Please note: you do not need to submit your blood glucose records every week to the team, rather, submit prior to scheduled visits, or if you notice you start to have patterns of highs - 3-4 blood sugar results are higher than normal at the same time (e.g. 3-4 out of 7 readings after dinner are high).

GDM Team Contact Information

The GDM team can be contacted by email or phone.

Email: gdm.program@sinaihealth.ca

Phone: 416-586-4800 extension 8626

Send a photo or digital copy of your blood sugar records to gdm.program@sinaihealth.ca on Sunday evening (4 days after introduction class).